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Chapter 13

Reimagining Compliance: Explainable AI Models for Financial Regulatory Audits

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ABSTRACT

This chapter develops an analytical model for Explainable AI (XAI) designed to support major types of banking regulatory compliance audits. Financial institutions increasingly rely on AI for efficiency in compliance tasks, but the opacity of complex models creates challenges in meeting regulatory transparency requirements. The proposed mathematical framework incorporates explainability as a quantifiable constraint in the optimization of AI models across five critical audit categories: CAMELS examinations, BSA/AML compliance, Consumer Compliance, IT Audits, and Internal Audits. By quantifying explainability through information-theoretic measures and Shapley values, the model balances predictive performance with regulatory requirements. The methodology is validated through a numerical example demonstrating significant improvement in the objective function while satisfying all regulatory constraints. The chapter addresses implementation challenges across different regulatory jurisdictions, human-AI collaboration considerations, and resource requirements for institutions of varying sizes.

1. INTRODUCTION

Banks operate under rigorous regulatory scrutiny, with compliance audits serving as important mechanisms to ensure sound practices (Ghosh et al., 2024). Traditional compliance methods, while effective, are increasingly supplemented by AI-driven systems for transaction monitoring, risk scoring, and data analysis (Doumpos et al., 2023). The increasing reliance on artificial intelligence (AI) to increase efficiency and decision-making introduces a key challenge: explainability. AI models, particularly complex ones like deep neural networks, often function as “black boxes,” making it difficult to understand or justify their decisions, which is a significant limitation in the context of regulatory compliance (Hassija et al., 2024). Thus, the opacity of these systems poses challenges in satisfying the explainability and transparency requirements demanded by regulators (Fernández, 2024; Torrie & Payette, 2023). In this chapter, I develop an analytical model for Explainable AI (XAI) designed to support major types of

DOI: 10.4018/979-8-3373-0209-6.ch013

regulatory compliance audits. Thus, this study addresses the fundamental question: *how can financial institutions balance AI's efficiency gains with regulatory requirements for explainability?*

Prior research has explored various aspects of AI explainability in financial services. Information-theoretic measures have been proposed to quantify relationships between model inputs and outputs (Islam et al., 2021; Piras et al., 2023), while Shapley values from cooperative game theory have been used to attribute importance to individual features (Holzinger et al., 2022). However, existing frameworks often lack integration of domain-specific regulatory constraints into model design (Longo et al., 2024). Current approaches tend to prioritize either accuracy or explainability but seldom balance these objectives within compliance audit contexts (Hosain et al., 2024).

This study contributes to the literature by developing an analytical model for Explainable AI (XAI) designed to support major types of regulatory compliance audits. The model incorporates constraints for five critical audit categories: Safety and Soundness Examinations (CAMELS System), Bank Secrecy Act/Anti-Money Laundering (BSA/AML) Compliance, Consumer Compliance (TILA, ECOA, CRA), Information Technology Audits, and Internal Audits. The model aims to provide a structured approach to balance predictive performance with regulatory transparency requirements. The methodology employs an optimization framework that maximizes model accuracy while incorporating explainability as a quantifiable constraint. The framework is validated through a numerical example using synthetic data, which demonstrates improvement in the objective function from -5.63 to 1.12 while satisfying all regulatory constraints.

2. THEORY

2.1 Major Types of Bank Compliance Audits

I focus on the following five main bank regulatory compliance audits, namely, (i) Safety and Soundness Examinations (CAMELS System), (ii) Bank Secrecy Act/Anti-Money Laundering (BSA/AML) Compliance, (iii) Consumer Compliance (TILA, ECOA, CRA), (iv) Information Technology (Cybersecurity, Data Protection), and (v) Internal Audits (Internal Control Reviews). Each compliance domain imposes unique constraints on explainability, which the model incorporates into its optimization framework.

2.1.1 Safety and Soundness Examinations (CAMELS System)

Safety and Soundness Examinations assess the ability of banks to withstand financial shocks, manage operational risks, and comply with regulatory mandates. Central to this process is the CAMELS rating system, a comprehensive framework used by regulatory agencies such as the Federal Reserve, Office of the Comptroller of the Currency (OCC), and the Federal Deposit Insurance Corporation (FDIC). The CAMELS system evaluates six critical dimensions of a bank's operations: Capital Adequacy, Asset Quality, Management Quality, Earnings, Liquidity, and Sensitivity to Market Risk (Kočenda & Iwasaki, 2022). Each component provides a lens through which regulators gauge a bank's overall health, assign risk ratings, and determine necessary supervisory actions.

The Capital adequacy component evaluates whether the bank maintains sufficient capital reserves to support its risk-weighted assets, as defined by international regulatory frameworks such as Basel III (Li et al., 2016). Regulators assess metrics like the Common Equity Tier 1 (CET1) capital ratio, Total

Capital Ratio, and Leverage Ratio (Le et al., 2023). The asset quality component focuses on the bank's loan and investment portfolios and evaluates their soundness and vulnerability to potential losses. Poor asset quality is a significant indicator of credit risk, often stemming from non-performing loans (NPLs) or investments in volatile or low-quality securities. Examiners may assess the level of delinquent or defaulted loans, the adequacy of loan loss reserves, and the concentration risk posed by sectoral or geographic dependencies (Badunenko et al., 2022). Management quality is a determinant of a bank's ability to identify, measure, and control risks effectively. Examiners assess the competence, experience, and strategic vision of the bank's leadership team, along with their responsiveness to emerging risks and regulatory requirements (Quoc Trung, 2021). Earnings stability and sustainability are essential for supporting a bank's operations, capital accumulation, and future growth. Examiners analyze financial performance metrics such as Net Interest Margin (NIM), Return on Assets (ROA), and Return on Equity (ROE). They also assess the quality of earnings, i.e., whether they arise from core banking operations or are reliant on one-time gains or volatile market activities (Al-Najjar & Assous, 2021).

Liquidity indicates a bank's ability to meet its financial obligations as they come due, without incurring significant losses (Akhtar et al., 2024). Examiners evaluate liquid asset holdings, the maturity profile of liabilities, and access to reliable funding sources. Metrics such as the Liquidity Coverage Ratio (LCR) and Net Stable Funding Ratio (NSFR), mandated under Basel III, play a key role in this assessment (Ben Lahouel et al., 2024). The Sensitivity to Market Risk component assesses the bank's exposure to market fluctuations, including changes in interest rates, foreign exchange rates, and commodity prices (Elyasiani, & Mansur, 2005; Handorf, 2016).

The CAMELS examination process is comprehensive, involving both quantitative and qualitative evaluations (Baer, 2024). The steps typically include:

1. *Data Collection:* Examiners gather extensive financial and operational data, including call reports, stress test results, and internal audit findings (Office of the Comptroller of the Currency (2022)).
2. *Risk Assessments:* AI and machine learning models are increasingly used to identify emerging risks, such as deteriorating asset quality or liquidity mismatches, through predictive analytics (Leo et al., 2019; Guerra & Castelli, 2021).
3. *Component Ratings:* Each CAMELS component is rated on a scale from 1 (strong performance) to 5 (critically deficient). The individual ratings are then combined to produce an overall composite score.
4. *Supervisory Actions:* Based on the composite rating, regulators determine whether the institution requires increased oversight, corrective action, or enforcement measures.

CAMELS ratings are confidential to encourage transparency during examinations and to maintain depositor confidence (Klein, 2020). Banks with strong ratings (1 or 2) are deemed to operate safely, while those with lower ratings (4 or 5) may face supervisory interventions, such as restrictions on growth, mandated capital injections, or even resolution proceedings in extreme cases (Stackhouse, 2018).

2.1.2 Bank Secrecy Act/Anti-Money Laundering (BSA/AML) Compliance Audits

The Bank Secrecy Act (BSA) of 1970, and subsequent anti-money laundering (AML) regulations form the centerpiece of the United States' financial crime prevention framework. These regulations are designed to prevent, detect, and report money laundering, terrorist financing, and other illicit financial

activities. BSA/AML compliance is an important obligation for financial institutions, and it requires them to implement robust systems and processes to monitor transactions, identify suspicious activities, and report them to regulatory authorities (FIFEC, 2021). Compliance audits for BSA/AML are conducted to ensure that financial institutions adhere to these regulatory requirements (FFIEC, 2021; Lessambo, 2023).

BSA/AML compliance audits are designed to (Miller & Kohr, 2017) (i) evaluate the effectiveness of a financial institution's anti-money laundering program; (ii) ensure adherence to recordkeeping and reporting requirements mandated under the BSA; (iii) detect gaps in customer due diligence (CDD), suspicious activity monitoring, and transaction reporting; (iv) mitigate risks associated with exposure to high-risk customers, jurisdictions, or products; and (v) assess the institution's ability to adapt to evolving financial crime tactics and regulatory expectations.

The core components of a BSA/AML compliance audit typically include the following:

- 1) *Customer Identification Program (CIP) and Customer Due Diligence (CDD)*: Financial institutions are required to verify the identity of their customers and assess their risk profile as part of their CIP and CDD processes (FFIEC, 2021).
- 2) *Suspicious Activity Monitoring and Reporting (SAR)*: Financial institutions are required to file Suspicious Activity Reports (SARs) when red flags are identified. Auditors evaluate (a) the effectiveness of the institution's transaction monitoring systems in identifying unusual patterns, such as structuring, rapid movement of funds, or transactions involving high-risk jurisdictions; (b) the timeliness and accuracy of SAR filings, which must be submitted to Financial Crimes Enforcement Network (FinCEN) within 30 calendar days of detecting suspicious activity; and (c) whether flagged activities are sufficiently documented and supported with evidence in the SAR (FFIEC, 2021).
- 3) *Currency Transaction Reporting (CTR)*: Financial institutions must file Currency Transaction Reports (CTRs) for cash transactions exceeding \$10,000 in a single day (FFIEC, 2021).
- 4) *Recordkeeping Requirements*: Under the BSA, financial institutions must retain records of customer transactions for at least five years (FFIEC, 2021). Auditors review (a) the completeness and accuracy of transaction logs, including wire transfers, deposits, and withdrawals, and (b) the institution's ability to retrieve records promptly upon request by regulators or law enforcement agencies.
- 5) *Independent Testing and Internal Audits*: The BSA requires financial institutions to conduct independent testing of their AML programs (FFIEC, 2021).
- 6) *Training Programs*: A robust BSA/AML program relies on ongoing training for employees and ensures they are equipped to identify and report suspicious activities (Howard, 2017; FFIEC, 2021). Auditors evaluate (a) the comprehensiveness of the training program, including content tailored to different roles (e.g., frontline staff, compliance officers); and (b) documentation of training attendance and frequency.

Modern financial institutions increasingly rely on AI and machine learning models to enhance BSA/AML compliance (Gupta et al., 2023). These tools analyze vast volumes of transactional data to detect anomalies indicative of financial crime. However, the "black box" nature of many AI models poses challenges in regulatory audits. Examiners focus on (a) Explainability (the ability of the institution to explain how the AI system identifies suspicious transactions or assigns risk ratings), (b) False Positives (the system's efficiency in reducing false positives, which can overwhelm compliance teams and divert resources from genuine threats), and (c) Scalability (the institution's ability to adapt monitoring systems to changes in transaction volumes, customer profiles, or emerging threats) (Estrada, 2020; Kute et al., 2021).

2.1.3 Consumer Compliance (TILA, ECOA, CRA) Audits

Consumer compliance audits are an important component of the regulatory framework that ensures financial institutions operate in a fair, transparent, and non-discriminatory manner when providing services to individuals and communities. These audits evaluate adherence to consumer protection laws such as the Truth in Lending Act (TILA), Equal Credit Opportunity Act (ECOA), and Community Reinvestment Act (CRA) (Nicoletti & Zhu, 2023). As regulators such as the Consumer Financial Protection Bureau (CFPB) and other federal banking agencies enforce these laws, consumer compliance audits serve as a mechanism for identifying violations, assessing systemic risks, and protecting consumer rights.

Examiners analyze credit applications, underwriting decisions, and pricing policies to detect disparities. Advanced analytical tools, including AI, are often employed to identify patterns of discrimination in credit approvals or terms (Griffith, 2023; Langenbucher, 2020; Ratzan & Rahman, 2023). Non-compliance with these laws can result in severe consequences, including substantial civil penalties, loss of public trust, and reputational harm. For example, the CRA aims to reduce the practice of “redlining” (i.e., denying credit to LMI communities) and promote equitable access to credit. CRA compliance audits assess a bank’s record of meeting community credit needs (Cyree & Winters, 2023). Examiners review lending data, community development activities, and outreach efforts (Avramidis et al., 2022; An et al., 2024). Poor CRA performance can result in restrictions on mergers, acquisitions, and branch expansions. It can also lead to reputational harm, as CRA ratings are publicly disclosed.

Consumer compliance audits for TILA, ECOA, and CRA typically involve the following steps (CFPB, 2015, 2023; FDIC, 2024a):

- 1) Examiners collect documentation such as loan applications, credit reports, and disclosures. Statistical models are used to analyze trends in lending decisions, pricing, and geographic distribution.
- 2) High-risk areas, such as subprime lending or redlining practices, are prioritized. AI and machine learning tools are increasingly used to identify anomalies or patterns indicative of non-compliance (Mostaghim, 2021).
- 3) Auditors conduct interviews with bank management and staff to understand policies and procedures. A review of internal compliance programs, including training and internal audit findings, is conducted.
- 4) Examiners provide a detailed report outlining findings, including any violations or deficiencies. Recommendations for corrective actions, such as policy revisions or enhanced employee training, are issued.

2.1.4 Information Technology (IT) Audits

With the financial sector increasingly targeted by cyberattacks and data breaches, regulatory bodies have established stringent requirements for cybersecurity, data protection, and IT governance (D'Onza et al., 2015). IT audits assess a financial institution’s adherence to these standards and focus on the adequacy of controls, risk management practices, and disaster recovery capabilities (Van Greuning & Bratanovic, 2020). Regulatory frameworks, such as the Federal Financial Institutions Examination Council (FFIEC) IT Examination Handbook, Gramm-Leach-Bliley Act (GLBA), and state-specific data protection laws, form the foundation of these audits (FFIEC, 2024; Nzeako et al., 2024).

IT audits aim to (FFIEC, 2024):

1. Assess the effectiveness of a bank's cybersecurity measures in safeguarding systems, networks, and data from unauthorized access and attacks.
2. Evaluate compliance with regulatory mandates for data privacy, protection, and secure processing.
3. Ensure the institution has robust disaster recovery and business continuity plans to minimize disruptions during crises.
4. Identify vulnerabilities in IT infrastructure and recommend corrective actions to mitigate risks.

IT compliance audits utilize advanced tools for threat detection, log analysis, and anomaly identification (FFIEC, 2024; KPMG, 2024). Examiners rely on standards and best practices from frameworks like International Organization for Standardization or ISO 27001, National Institute of Standards and Technology or NIST, and Control Objectives for Information and Related Technologies or COBIT to benchmark IT governance and security practices (Al Hadad & Maulana, 2023). They also review banks' adherence to privacy laws, such as the GLBA, California Consumer Privacy Act (CCPA), and European Union's General Data Protection Regulation (GDPR), if applicable (Davis, 2020).

2.1.5 Internal Audits

Internal audits are an integral component of a bank's governance framework, which ensure the effectiveness of internal controls, risk management practices, and overall compliance with regulatory requirements (Jarrah et al., 2022). These audits, conducted by the bank's internal audit department or outsourced to third-party firms, are mandated by regulatory agencies such as the Federal Reserve, the Office of the Comptroller of the Currency (OCC), and the Federal Deposit Insurance Corporation (FDIC) (FDIC, 2024b). They primarily report findings to internal stakeholders like the audit committee and management with a focus on improving internal processes. Internal audits serve as a first line of defense in identifying and mitigating risks across operational, financial, and technological domains.

The primary objectives of internal audits in the banking sector include (FDIC, 2024c):

- 1) Assess whether the bank's internal control systems adequately mitigate risks related to fraud, financial misstatements, and operational inefficiencies.
- 2) Verify adherence to laws and regulations such as the Sarbanes-Oxley Act (SOX), the Gramm-Leach-Bliley Act (GLBA), and anti-money laundering (AML) provisions.
- 3) Identify gaps in risk management frameworks and recommend corrective actions.
- 4) Highlight inefficiencies in processes and recommend enhancements to improve productivity and cost-effectiveness.
- 5) Provide insights to the board of directors and audit committees to support strategic decision-making and oversight.

There are several overlaps in the steps for doing internal audits and the steps for doing the compliance audits described in the earlier sections, particularly in terms of planning, risk assessment, data collection, reviewing documentation, testing controls, analyzing findings, reporting results and recommending corrective actions. However, internal audits are broader in scope and focus on the bank's internal control systems, governance frameworks, and operational efficiency. Other compliance audits, such as BSA/AML, TILA, or CAMELS examinations, are more targeted and address specific areas like financial stability, anti-money laundering, or consumer protection.

2.2 Explainable AI in Financial Services

Artificial intelligence (AI) in financial services has enabled automation, efficiency, and precision in decision-making processes such as credit scoring, fraud detection, and compliance audits (Černevičienė & Kabašinskas, 2024). However, the opaque nature of many AI models, particularly deep learning algorithms, has raised significant concerns about their interpretability and accountability. This has led to an increasing focus on Explainable AI (XAI) to address the need for transparency in financial applications. AI models used in financial services often operate as “black boxes” and produce outputs without providing intuitive explanations for their decisions (Fernández, 2024). This lack of transparency is particularly problematic in highly regulated sectors such as banking, where decisions must be explainable to auditors, regulators, and affected stakeholders. For example, anti-money laundering (AML) systems require the ability to justify flagged transactions, and credit scoring models must provide fair and interpretable evaluations to avoid biases against protected groups (Giudici & Raffinetti, 2021).

The European Union’s General Data Protection Regulation (GDPR) has been particularly influential, with Articles 13-15 establishing a “right to explanation” for automated decisions affecting individuals (Kaminski, 2019). This regulatory requirement has accelerated the development of post-hoc explainability methods for black-box models. Similarly, the United States’ Fair Credit Reporting Act (FCRA) and Equal Credit Opportunity Act (ECOA) require that adverse credit decisions provide specific reasons, which indirectly mandates some degree of explainability in lending algorithms (Selbst & Barocas, 2018). The historical evolution of XAI in banking compliance reflects the tension between technological advancement and regulatory oversight. Early approaches to explainable financial models in the 1990s and 2000s emphasized rule-based systems and decision trees, which offered transparency at the expense of predictive power (Arrieta et al., 2020).

Cross-jurisdictional challenges further complicate the implementation of XAI in global banking operations. Financial institutions operating across multiple regions must reconcile differing standards for explainability, from the principles-based approach of the EU to the more rules-based frameworks in the United States and Singapore (Floridi et al., 2018). The Basel Committee on Banking Supervision has recognized this challenge and indicated the need for “explainability by design” in its principles for operational resilience, though concrete standards remain underdeveloped (BIS, 2021).

Explainable AI seeks to bridge this gap by providing techniques and frameworks to make AI decisions interpretable. Two prominent approaches include:

1. *Information-Theoretic Measures:* Metrics like mutual information quantify the relationship between input features and outputs. They provide insights into the importance of specific variables in a model’s predictions (Islam et al., 2021; Piras et al., 2023). In financial applications, mutual information can be used to determine how features like credit history or transaction patterns contribute to loan approvals or fraud detection.
2. *Shapley Values:* Derived from cooperative game theory, Shapley values attribute contributions to individual input features by analyzing their marginal impact on a model’s output (Holzinger et al., 2022). This method is particularly effective in identifying feature importance and ensuring fairness in credit scoring and lending decisions (Giudici et al., 2024).

Traditional AI models often struggle to meet these requirements, which leads to a gap between operational performance and regulatory compliance. Existing approaches to XAI in financial compliance have several notable limitations. First, many frameworks treat explainability as a post-hoc consideration rather than an intrinsic design parameter, which leads to models that may satisfy technical transparency requirements without providing meaningful insights to non-technical stakeholders like regulators or customers (Carvalho et al., 2019). Second, current methodologies often fail to address the trade-off between global and local explainability. A model may provide accurate explanations for individual decisions while obscuring systemic patterns that could indicate discriminatory practices or other compliance risks (Rudin, 2019). Third, the computational complexity of calculating certain explainability metrics, such as Shapley values for high-dimensional data, creates practical challenges for real-time compliance monitoring and reporting (Molnar, 2020). Current approaches prioritize either accuracy or explainability but seldom balance these objectives within the context of compliance audits (Longo et al., 2024). This highlights the need for a mathematical framework that explicitly incorporates explainability as a constraint within the optimization process and balances predictive performance, regulatory compliance, and interpretability.

3. MATHEMATICAL MODEL

The absence of explainability in many AI-driven systems poses significant risks in satisfying the stringent requirements of regulatory bodies. To address this, I propose a mathematical model that incorporates explainability as a quantifiable constraint within the optimization framework of AI models, which can ensure compliance with regulatory mandates while maintaining operational efficiency. The model is tailored for the five major categories of bank compliance audits discussed in the prior section. Each category imposes unique constraints and requirements, which the model accommodates.

3.1 Model Framework

The objective is to maximize the predictive accuracy A of AI models used in compliance tasks while incorporating explainability E and ensuring regulatory compliance. The optimization problem is expressed as:

$$\max_M A(M) - \lambda \cdot C(M) + \gamma \cdot E(M)$$

Where M indicate the AI model parameters; $A(M)$ indicates predictive accuracy of the model; $C(M)$ indicates complexity of the model; $E(M)$ indicates explainability score; λ is the weighting factor for model complexity; and γ is the weighting factor for model explainability. The AI model parameters (M) are the adjustable variables or weights within a machine learning or artificial intelligence model (e.g., weights in a neural network) that are learned from training data during the model's training phase. These parameters govern the behavior of the model and determine how it processes input data and produces predictions or outputs. Thus, this objective function balances accuracy and explainability while penalizing excessive complexity, which may reduce interpretability.

The explainability score $E(M)$ is quantified using information-theoretic measures, such as mutual information and Shapley values. The constraint below ensures that the model meets a minimum explainability threshold E_{min} required by regulators.

$$E(M) \geq E_{min}$$

Mutual Information (MI) is a metric that quantifies the amount of information shared between two random variables, i.e., it measures how much knowing one variable reduces the uncertainty about the other (Islam et al., 2021; Piras et al., 2023). For an AI model, MI can be used to evaluate how much the input features (e.g., loan amount, credit score) explain or contribute to the output (e.g., loan approval). MI thus helps identify which input features contribute the most to the model's predictions by quantifying the relationship between features and outputs (Li et al., 2020; Nauta et al., 2023). Features with higher MI scores provide more information about the output, which makes the model's decisions more interpretable. On the other hand, Shapley values come from cooperative game theory and are used to fairly distribute the "payout" (in this case, the model's output) among all input features based on their contribution to the prediction (Holzinger et al., 2022). They assign each feature an importance score by considering all possible combinations of feature contributions (Giudici & Raffinetti, 2021). Shapley values thus provide a way to determine each feature's contribution to a specific prediction and ensure fairness by considering every possible interaction between features.

Now, each compliance audit category imposes specific constraints as follows:

- 1) *CAMELS*: Ensure interpretability of risk weight assignments, stress-test outcomes, and CAMELS ratings.
- 2) *BSA/AML*: Minimize false positives in suspicious activity detection while maintaining explainable transaction monitoring models.
- 3) *Consumer Compliance*: Ensure bias mitigation in AI-driven credit scoring models, with clear feature attributions for decisions.
- 4) *IT Audits*: Guarantee transparency in cybersecurity anomaly detection and system risk assessments.
- 5) *Internal Audits*: Provide detailed justifications for flagged control deficiencies.

These constraints are incorporated into the optimization framework as additional terms or bounds. These constraints enforce adherence to regulatory requirements while maintaining explainability and model performance.

The mathematical representation of the CAMELS Compliance constraint is given below:

$$I(X; Y) \geq I_{min}$$

Where, $I(X; Y)$ indicates mutual information between features X (e.g., risk weights, liquidity ratios) and outputs Y (e.g., CAMELS ratings); and I_{min} indicates the minimum mutual information threshold that ensures adequate interpretability. This ensures that the features used in the model (e.g., capital adequacy, asset quality) provide sufficient explanatory power for the assigned ratings. Thus, the CAMELS constraint ensures that the model's outputs (e.g., CAMELS ratings, stress-test results) are interpretable in terms of feature contributions.

The mathematical representation of the BSA/AML Compliance constraint is given below:

$$FP \leq FP_{max} \text{ and } E(M) \geq E_{min}$$

Where FP indicates the number of false positives in flagged suspicious transactions; FP_{max} indicates the maximum allowable false positives; $E(M)$ indicates the explainability score of the model (e.g., based on Shapley values or feature importance); and E_{min} is the minimum explainability threshold. This dual BSA/AML constraint ensures the model minimizes false positives, which overwhelm compliance teams, while remaining interpretable enough to justify flagged transactions. Thus, the BSA/ AML constraint balances the trade-off between reducing false positives and ensuring explainability in flagged transactions.

The mathematical representation of the Consumer Compliance constraint is given below:

$$\Delta\text{Approval} \leq \epsilon \text{ and } \phi_i \geq \phi_{min}$$

Where, $\Delta\text{Approval}$ is the difference in approval rates between protected and unprotected groups; ϵ indicates the acceptable threshold for disparity; ϕ_i indicates the Shapley value for feature i , which represents its contribution to the model's decisions; and ϕ_{min} is the minimum attribution threshold or the minimum acceptable contribution or importance assigned to key input features (e.g., income, credit history). ϕ_{min} is thus the minimum value that a key feature's Shapley attribution score must meet or exceed to ensure it is adequately considered by the model in its decision-making process. This threshold ensures that important features are not underrepresented in the decision-making process in regulated applications such as lending. Thus, this constraint helps check that credit decisions are fair (low disparity) and interpretable (clear feature contributions). It ensures fairness in these decisions by eliminating bias against protected groups while providing clear feature attributions.

The mathematical representation of the IT Audit constraint is given below:

$$P_{critical} \geq P_{min} \text{ and } H(X) - H(X|Y) \geq T_{min}$$

Where, $P_{critical}$ is the probability of correctly detecting a critical anomaly; P_{min} is the minimum detection probability required by regulators; $H(X) - H(X|Y)$ indicates mutual information between input features X and model outputs Y ; and T_{min} is the mutual information required to satisfy transparency standards. This ensures the cybersecurity system is both effective and explainable, which allows auditors to understand why specific anomalies were flagged.

Here, X would represent input features (e.g., login time, IP address, data volume) and Y model outputs (e.g., anomaly flag). Mutual information, $I(X; Y)$, quantifies the amount of information that the input features X provide about the output Y , i.e., it measures how much knowing X reduces uncertainty about Y (Ayoub et al., 2022). It is derived from the difference between $H(X)$, the entropy of X , which represents the uncertainty or randomness in the input features, and $H(X|Y)$, the conditional entropy of X given Y , which represents the remaining uncertainty in X after knowing Y (Ayoub et al., 2022). T_{min} is the minimum information threshold, which ensures that there is a sufficient relationship between the inputs and outputs for the model to be considered interpretable.

The equation $H(X) - H(X|Y) \geq T_{min}$ ensures that the amount of information X provides about Y is above a predefined minimum threshold T_{min} , which would indicate that the model is sufficiently transparent. This also builds confidence that the model does not rely on opaque or arbitrary relationships but instead uses clear, quantifiable dependencies between the inputs and outputs. A high value of $I(X; Y)$ or $H(X) - H(X|Y)$ indicates that the model's output Y can be well-explained by the input features X . If $I(X; Y)$ is too low, the model might be relying on hidden patterns or noise, which makes it difficult to

justify its decisions. Thus, this constraint guarantees transparency in anomaly detection while ensuring the system identifies risks effectively.

The mathematical representation of the Internal Audit constraint is given below:

$$\text{Coverage}(D) \geq C_{\min} \text{ and } E(M) \geq E_{\min}$$

Where, $\text{Coverage}(D)$ is the proportion of flagged deficiencies D for which clear justifications are provided; $C_{\min}$ is the minimum required proportion of flagged issues that must have explanations; $E(M)$ is the explainability score of the model's output; and $E_{\min}$ represents the minimum explainability threshold required for an AI model M to satisfy transparency and interpretability standards in regulatory compliance settings, i.e., it is a predefined benchmark that ensures the model's decisions can be adequately explained, either to internal stakeholders or external regulators.

Here, $\text{Coverage}(D)$ could indicate issues in internal controls or anomalies that are accompanied by clear justifications or explanations. The equation $\text{Coverage}(D) \geq C_{\min}$ constraint ensures that most, if not all, flagged deficiencies are justified with clear, auditable explanations. $E(M)$ quantifies how well the model's decisions can be understood in terms of input features and outputs. The equation $E(M) \geq E_{\min}$ ensures that these attributions meet the minimum transparency standard. Together, these constraints build confidence that flagged deficiencies in internal audits are well-justified and clearly explained to stakeholders.

To sum it up, the regulatory constraints are incorporated into the overall optimization problem as either hard constraints or penalty terms as follows:

$$\max_M A(M) - \lambda \cdot C(M) + \gamma \cdot E(M)$$

Subject to:

$$I(X; Y) \geq I_{\min} \text{ (CAMELS Compliance)}$$

$$FP \leq FP_{\max} \text{ and } E(M) \geq E_{\min} \text{ (BSA/AML Compliance)}$$

$$\Delta \text{Approval} \leq \epsilon \text{ and } \phi_i \geq \phi_{\min} \text{ (Consumer Compliance)}$$

$$P_{\text{critical}} \geq P_{\min} \text{ and } H(X) - H(X|Y) \geq T_{\min} \text{ (IT Audits)}$$

$$\text{Coverage}(D) \geq C_{\min} \text{ and } E(M) \geq E_{\min} \text{ (Internal Audits)}$$

3.2 Practical Integration into Compliance Workflows

While the mathematical formulation establishes a theoretical foundation for explainable AI in banking compliance, successful implementation requires integration into existing compliance workflows. Explainability should be considered at each stage of the compliance process. Table 1 illustrates how the analytical framework translates into practical workflows at each stage of the compliance process across different compliance domains.

Table 1. End-to-end XAI integration in banking compliance workflows

Compliance Domain	Input	Processing	Explanation Generation	Compliance Documentation	Regulatory Defense/Engagement
BSA/AML Transaction Monitoring	Customer transaction data	Model flags suspicious activities while maintaining false positive rate below threshold	Calculates Shapley values to quantify contribution of each feature (transaction amount, frequency, customer risk score, etc.)	Automatically generates SAR narratives incorporating key factors from explainability analysis	Demonstrates that flagged transactions meet both detection accuracy and explainability requirements during examinations
Consumer Compliance Fair Lending	Loan application data	Evaluates creditworthiness while applying fairness constraint and feature attribution requirements	Generates counterfactual explanations identifying minimum changes needed for approval	Translates explanations into compliant adverse action notices with clear reasons for denial	Uses aggregated explanation data for comparative analyses across demographic groups
CAMELS Financial Stability Assessment	Financial performance metrics, risk measures, governance data	Generates component scores while maintaining required mutual information between inputs and outputs	Produces hierarchical explanations showing how specific financial metrics influence each CAMELS component	Provides supporting documentation for rating determinations	Enables officials to discuss rating determinants with examiners and develop targeted improvement plans for lower-rated components

4. NUMERICAL EXAMPLE

In this section, I provide a numerical example for the optimization problem with explainability and compliance constraints. Let's consider a bank deploying an AI-based system for anomaly detection in transaction monitoring. The system aims to optimize detection performance while satisfying regulatory explainability requirements. The optimization problem incorporates constraints for different compliance audits: CAMELS (safety and soundness), BSA/AML (anti-money laundering), Consumer Compliance (fair lending practices), IT Audits (cybersecurity), and Internal Audits (internal control justifications). This example demonstrates the model's application, using synthetic data to evaluate how the constraints affect the optimization process. The goal is to maximize the model's predictive accuracy $A(M)$, which is balanced with complexity $C(M)$ and explainability $E(M)$.

For CAMELS Compliance, mutual information $I(X; Y)$ between input features X (e.g., risk weights, asset quality) and model outputs Y (e.g., CAMELS rating) must meet the minimum threshold as follows:

$$I(X; Y) \geq I_{\min} = 1.5 \text{ bits}$$

A bit is the basic unit of information in information theory, and bits are used to measure the amount of uncertainty, information, or entropy in a system (Witten, 2020). Higher mutual information (in bits) typically means the model's decisions are more explainable.

For BSA/AML Compliance, false positives in suspicious transaction detection must be below a maximum allowable rate, and the explainability score $E(M)$ must meet a minimum threshold as follows:

$$FP \leq FP_{max} = 5\% \text{ and } E(M) \geq E_{min} = 0.8$$

For Consumer Compliance, the disparity in loan approval rates between protected and unprotected groups must not exceed a certain threshold rate, and key feature attribution scores ϕ_i must meet a minimum threshold as follows:

$$\Delta\text{Approval} \leq \epsilon = 0.05 \text{ and } \phi_i \geq \phi_{min} = 0.2$$

For IT Audits, anomaly detection must meet a minimum detection probability, and mutual information between inputs X (e.g., login time, IP address) and outputs Y (anomaly flags) must meet a transparency threshold as follows:

$$P_{critical} \geq 90\% \text{ and } H(X) - H(X|Y) \geq T_{min} = 1.0 \text{ bit}$$

For Internal Audits, proportion of flagged deficiencies D with justifications must be above a threshold, and explainability $E(M)$ must meet a minimum level as follows:

$$\text{Coverage}(D) \geq C_{min} = 95\% \text{ and } E(M) \geq E_{min} = 0.8\%$$

Thus, the optimization problem in this illustrative scenario with the synthetic data values is as follows:

$$\max_M A(M) - \lambda \cdot C(M) + \gamma \cdot E(M)$$

Subject to: $I(X; Y) \geq 1.5$ bits; $FP \leq 5\%$; $E(M) \geq 0.8$; $\Delta\text{Approval} \leq 0.05$; $\phi_i \geq 0.2$; $P_{critical} \geq 90\%$; $H(X) - H(X|Y) \geq 1.0$ bits; and $\text{Coverage}(D) \geq 95\%$.

The optimization problem is transformed into a constrained optimization problem by incorporating penalties for any violations of constraints directly into the objective function. This helps to solve the problem as an unconstrained optimization, where penalties discourage the solution from violating the constraints. Each constraint is thus represented as a penalty term. If a constraint is violated, the penalty increases the cost, reducing the overall objective value. Let α_i represent the weight (penalty factor) associated with each constraint. Then,

For CAMELS Compliance,

$$\text{Penalty: } -\alpha_1 \cdot \max(0, 1.5 - I(X; Y))$$

For BSA/AML Compliance,

$$\text{Penalties: } -\alpha_2 \cdot \max(0, FP - 0.05) \text{ and } -\alpha_3 \cdot \max(0, 0.8 - E(M))$$

For Consumer Compliance,

Penalties: $-\alpha_4 \cdot \max(0, \Delta\text{Approval} - 0.05)$ and $-\alpha_5 \cdot \sum_i \max(0, 0.2 - \phi_i)$, which is summed for all critical features i .

For IT Audits,

Penalties: $-\alpha_6 \cdot \max(0, 0.9 - P_{\text{critical}})$ and $-\alpha_7 \cdot \max(0, 1.0 - (H(X) - H(X|Y)))$.

For Internal Audits,

Penalty: $-\alpha_8 \cdot \max(0, 0.95 - \text{Coverage}(D))$

Thus, the final transformed or the full penalized objective function, becomes:

$$\begin{aligned} L(M) = & A(M) - \lambda \cdot C(M) + \gamma \cdot E(M) - \alpha_1 \cdot \max(0, 1.5 - I(X; Y)) - \alpha_2 \cdot \max(0, FP - 0.05) \\ & - \alpha_3 \cdot \max(0, 0.8 - E(M)) - \alpha_4 \cdot \max(0, \Delta\text{Approval} - 0.05) - \alpha_5 \cdot \sum_i \max(0, 0.2 - \phi_i) \\ & - \alpha_6 \cdot \max(0, 0.9 - P_{\text{critical}}) - \alpha_7 \cdot \max(0, 1.0 - (H(X) - H(X|Y))) - \alpha_8 \cdot \max(0, 0.95 - \text{Coverage}(D)) \end{aligned}$$

To reiterate, all the penalty terms ($\alpha_i \cdot \max$) reduce the objective value when constraints are violated.

Now, to solve the optimization problem further, a combination of gradient-based methods and sub-gradient approaches can be used. For differentiable components, such as the objective function terms $A(M)$, $C(M)$, and $E(M)$, stochastic gradient descent (SGD) could be employed to iteratively refine the model parameters M . Non-differentiable penalty terms, such as $\max(0, \cdot)$, can be handled using sub-gradient methods or smooth approximations to ensure convergence. The optimization would begin with initial parameters M_0 , where the objective function $L(M)$ is evaluated, and gradients or sub-gradients are calculated at each iteration. Through iterative refinement, the model parameters would need to be adjusted to maximize $L(M)$ so that accuracy and explainability is improved while reducing penalties associated with constraint violations. For example, adjustments might aim to reduce the false-positive rate (FP) and improve the explainability score ($E(M)$), which brings the model into compliance with requirements like $FP \leq 5\%$ and $E(M) \geq 0.8$. After the optimization process converges, the final model is validated by verifying that all constraints are satisfied.

For the purpose of this illustrative scenario, let's assume the following initial metric values of the AI model:

Accuracy - $A(M) = 94\%$; Complexity - $C(M) = 0.4$; Explainability - $E(M) = 0.7$; Mutual Information - $I(X; Y) = 1.2\text{bits}$; False-Positive Rate - $FP = 7\%$; Approval Disparity - $\Delta\text{Approval} = 0.08$; Feature Attribution - $\phi_i = 0.15$ (critical features like income, credit history); Critical Detection Probability - $P_{\text{critical}} = 88\%$; Transparency (IT) - $H(X) - H(X|Y) = 0.9\text{bits}$; and Justification Coverage - $\text{Coverage}(D) = 90\%$.

The weights for penalties are as follows: $\lambda = 0.2$, $\gamma = 0.3$, and $\alpha_1, \dots, \alpha_8$. Substituting these values into the full penalized objective function,

$$\begin{aligned} L(M) = & 0.94 - 0.2 \cdot 0.4 + 0.3 \cdot 0.7 - 10 \cdot \max(0, 1.5 - 1.2) - 10 \cdot \max(0, 0.07 - 0.05) - 10 \cdot \\ & \max(0, 0.8 - 0.7) - 10 \cdot \max(0, 0.08 - 0.05) - 10 \cdot \sum_i \max(0, 0.2 - 0.15) - 10 \cdot \max(0, 0.9 \\ & - 0.88) - 10 \cdot \max(0, 1.0 - 0.9) - 10 \cdot \max(0, 0.95 - 0.9) \end{aligned}$$

Thus, $L(M) = 0.94 - 0.08 + 0.21 - 3 - 0.2 - 1 - 0.3 - 0.5 - 0.2 - 1 - 0.5 = -5.63$.

After the optimization process, the AI model parameters M are adjusted to balance accuracy, complexity, and explainability while satisfying all regulatory constraints. This involves iterative refinement to improve the model's compliance performance and minimize penalties associated with constraint violations. Key outcomes after optimization include:

- 1) Accuracy ($A(M)$) may slightly decrease due to trade-offs required for enhancing transparency and fairness.
- 2) The complexity penalty ($C(M)$) is reduced to align with regulatory expectations while maintaining model performance.
- 3) The explainability score ($E(M)$) is improved to meet or exceed the minimum required threshold (E_{min}).

Also, all regulatory constraints are verified to be satisfied. For example:

- 1) The false-positive rate (FP) is reduced to within acceptable limits.
- 2) Mutual information ($I(X; Y)$) meets the CAMELS transparency requirement.
- 3) Approval disparities and feature attribution thresholds in Consumer Compliance are within permissible bounds.
- 4) Critical anomaly detection probability ($P_{critical}$) and IT transparency measures are improved.
- 5) Internal audit coverage ($Coverage(D)$) exceeds the required minimum.

Thus, after optimization, the refined metrics are as follows:

Accuracy - $A(M) = 93\%$; Complexity - $C(M) = 0.35$; Explainability - $E(M) = 0.85$; Mutual Information - $I(X; Y) = 1.6bits$; False-Positive Rate - $FP = 4.5\%$; Approval Disparity - $\Delta Approval = 0.04$; Feature Attribution - $\phi_i = 0.2$; Critical Detection Probability - $P_{critical} = 91\%$; Transparency (IT) - $H(X) - H(X|Y) = 1.1bits$; and Justification Coverage - $Coverage(D) = 96\%$.

Recalculating $L(M)$,

$$L(M) = 0.93 - 0.2 \cdot 0.35 + 0.3 \cdot 0.85 - 10 \cdot \max(0, 1.5 - 1.6) - 10 \cdot \max(0, 0.045 - 0.05) - 10 \cdot$$

$$\max(0, 0.8 - 0.85) - 10 \cdot \max(0, 0.04 - 0.05) - 10 \cdot \sum_i \max(0, 0.2 - 0.2) - 10 \cdot \max$$

$$(0, 0.9 - 0.91) - 10 \cdot \max(0, 1.0 - 1.1) - 10 \cdot \max(0, 0.95 - 0.96)$$

$$\text{Thus, } L(M) = 0.93 - 0.07 + 0.255 - 0 - 0 - 0 - 0 - 0 - 0 - 0 - 0 = 1.12.$$

To sum it up, after optimization, the objective function $L(M)$ improves from -5.63 to 1.12 , and all constraints are satisfied. The increase in the refined objective function indicates an optimal balance between accuracy, explainability, and regulatory compliance. The optimized model is practical for real world implementation so that effective performance is achieved while adhering to stringent regulatory requirements.

5. LIMITATIONS

Despite its practical applications and ability to integrate explainability into regulatory compliance frameworks, the proposed mathematical model has several limitations that warrant consideration.

First, the model quantifies explainability using mutual information and Shapley values. Mutual Information assumes that input-output relationships can be fully captured through probabilistic measures. This may oversimplify complex dependencies, particularly in high-dimensional datasets. Shapley Values attribute importance to individual features but can be computationally prohibitive for models with numerous input features or complex interdependencies.

Second, the optimization framework inherently involves trade-offs, where increasing explainability or reducing complexity may decrease predictive accuracy. While this is necessary to meet regulatory standards, it may limit the model's utility in scenarios where maximum predictive performance is important, such as fraud detection or high-frequency trading. Incorporating multiple regulatory constraints, each with its own penalty terms, significantly increases the computational burden of solving the optimization problem. For example, calculating Shapley values involves examining all possible combinations of input features, which grows exponentially with the number of features.

Third, a significant limitation of this study is the reliance on synthetic data rather than empirical validation with real-world financial datasets. Empirical validation would require access to proprietary banking data containing sensitive compliance information, transaction patterns, and historical regulatory outcomes. Such data is subject to stringent privacy regulations, confidentiality requirements, and competitive considerations that make it exceptionally difficult to obtain for research purposes. Financial institutions are generally unable to share transaction monitoring data, CAMELS ratings, or regulatory examination findings due to both legal restrictions and reputational risk concerns.

Fourth, the implementation of explainable AI models in banking compliance faces varying challenges across different regulatory jurisdictions. In the United States, principles-based approaches from agencies like the Federal Reserve and OCC emphasize outcomes and risk management rather than prescribing specific technical solutions (American Bankers Association, 2020). This contrasts with the European Union's more prescriptive approach under the AI Act, which categorizes AI systems for banking compliance as "high-risk" applications requiring robust transparency, documentation, and human oversight (European Commission, 2021). These divergent approaches create implementation challenges for global financial institutions. For example, a model that meets the explainability standards for BSA/AML compliance in the U.S. may not satisfy the "right to explanation" provisions under the EU's GDPR.

Fifth, the practical application of explainable AI in compliance audits depends critically on how human auditors interpret and act upon model outputs. While the mathematical model in this chapter quantifies explainability through metrics like mutual information and Shapley values, translating these technical measures into actionable insights for auditors remains challenging (Bhatt et al., 2020). Effective human-AI collaboration in compliance audits requires consideration of both the technical accuracy of explanations and their cognitive accessibility to users with varying technical backgrounds.

Finally, regulatory environments are dynamic, with frequent updates to compliance standards and expectations. The model's static constraints may require continuous re-calibration to remain compliant, which may necessitate additional resources and expertise. Banking regulations evolve in response to emerging risks, technological innovations, and changing market conditions. Future iterations of the framework should incorporate mechanisms for dynamically updating explainability constraints based on regulatory changes without requiring complete model retraining. One promising approach is to develop meta-learning

frameworks that can rapidly adapt to new regulatory contexts by leveraging knowledge from previously encountered compliance scenarios (Finn et al., 2017). Additionally, incorporating formal verification techniques could provide stronger guarantees about model behavior under changing regulatory conditions (Katz et al., 2017).

Case studies from different banking environments illustrate these challenges. For example, a large multinational bank implementing AI-based transaction monitoring systems across its global operations must navigate not only technical challenges in model explainability but also variations in how regulators across different jurisdictions interpret and evaluate those explanations. Similarly, a community bank adopting AI for credit scoring must ensure that explanations are accessible to both internal compliance teams and external examiners who may have limited technical expertise in machine learning. These practical considerations highlight the importance of designing explainability frameworks that are both technically robust and operationally feasible across diverse regulatory and organizational contexts.

6. REAL WORLD COMPLEXITIES IN COMPLIANCE

While the numerical example above demonstrates the model's application in a simplified setting, real-world banking compliance scenarios introduce additional complexities that merit consideration. In practice, financial institutions face multidimensional compliance requirements across heterogeneous datasets that vary in size, structure, and quality.

For example, a large bank's BSA/AML monitoring system may process millions of transactions daily across diverse channels (wire transfers, ACH, card payments, etc.), each with different risk profiles and data attributes. The transaction monitoring algorithm must synthesize this heterogeneous data while maintaining explainability across multiple compliance dimensions simultaneously. A suspicious transaction might trigger both BSA/AML and consumer compliance considerations. This may require coordinated explanations that satisfy different regulatory frameworks.

Also, in the real world, the BSA/AML monitoring system would ingest data from core banking systems, customer information files (CIF), watch lists, and external sources like Financial Intelligence Units (FIUs). Each source contributes different features with varying reliability and completeness. The explainability constraints would need to account for this variation through confidence intervals around metrics like mutual information ($I(X; Y)$) and feature attribution scores (ϕ_i). A transaction flagged as suspicious for BSA/AML purposes might simultaneously raise consumer compliance concerns if it relates to a credit decision affecting a protected class. The feasible solution space narrows considerably when multiple compliance frameworks must be satisfied simultaneously, which often requires more sophisticated model architectures or ensemble approaches.

The optimization outcomes can also vary significantly based on data quality and organizational context. Banks with more mature data governance frameworks would likely achieve higher post-optimization objective function values compared to those with fragmented data infrastructures. This indicates that data quality serves as a limiting factor for explainability performance. Explainability performance could also significantly vary across different regulatory domains. The BSA/AML models may achieve post-optimization objective function values that are different from consumer compliance models, which suggests domain-specific challenges in balancing accuracy and explainability.

The application of the proposed mathematical framework at enterprise scale also introduces significant computational and operational challenges. Many large financial institutions process billions of transactions annually across thousands of features, which makes direct calculation of metrics like Shapley values computationally prohibitive. Most banks also have substantial investments in legacy compliance systems that cannot be replaced wholesale. The optimization framework would need to interface with existing transaction monitoring, case management, and regulatory reporting systems. This integration typically requires (a) a layered architecture that applies the explainability constraints without disrupting core transaction processing; (b) API-based interfaces to existing compliance workflows; and (c) governance mechanisms to track and document model changes as explainability constraints evolve.

Financial institutions implementing such frameworks should adopt a phased approach, beginning with specific high-risk compliance domains before expanding to enterprise-wide deployment. They should also establish continuous monitoring mechanisms to detect potential degradation in explainability metrics over time as data distributions or regulatory expectations change.

The computational and expertise requirements for implementing the proposed mathematical framework present significant challenges, especially for smaller financial institutions with limited resources. Financial institutions vary dramatically in their capacity to implement sophisticated AI explainability frameworks. Large multinational banks typically possess dedicated data science teams, substantial computing infrastructure, and specialized compliance technology personnel. In contrast, community banks and smaller credit unions often rely on third-party vendors for compliance solutions and may lack specialized AI expertise (Turksen et al., 2024). This resource disparity creates implementation barriers that could exacerbate technological inequality in the financial sector. For institutions with limited in-house expertise, the calculation of complex explainability metrics like Shapley values and mutual information measures may be prohibitively difficult without external support. The Federal Financial Institutions Examination Council (FFIEC) has also recognized these resource disparities and issued guidance encouraging tailored approaches to model risk management based on an institution's size, complexity, and risk profile (FFIEC, 2023).

To address this issue, financial institutions can adopt a risk-based approach where explainability requirements are calibrated to both the complexity of the AI systems and the institution's size and risk profile. For smaller institutions with less complex models, simplified explainability metrics focused on feature importance and decision rules may be sufficient to satisfy regulatory requirements while remaining computationally tractable. Industry consortia or regulatory technology (RegTech) providers could develop shared platforms that implement the proposed framework as a service. Technologies like Apache Spark, Ray, or Dask enable distributed computation of explainability metrics across computing clusters and may make previously intractable calculations feasible even for smaller institutions by leveraging cloud resources on demand rather than maintaining expensive infrastructure. This would spread development and operational costs across multiple institutions. These platforms could provide standardized explainability metrics and optimization capabilities that smaller banks could use without building in-house expertise.

7. CONCLUSION

This study addressed the critical challenge of incorporating explainability into AI models used for banking regulatory compliance audits. The research focused on developing a mathematical framework that balances predictive accuracy with regulatory transparency requirements across five major types of compliance audits: CAMELS examinations, BSA/AML compliance, consumer compliance, IT audits, and internal audits. The proposed mathematical model successfully integrates explainability as a quantifiable constraint within the optimization framework through two key mechanisms: information-theoretic measures and Shapley values.

The model's formulation demonstrates how predictive accuracy can be optimized while maintaining regulatory compliance and model interpretability. The numerical example validated the model's effectiveness by showing improvement in the objective function from -5.63 to 1.12 while satisfying all regulatory constraints. This research contributes significantly to the intersection of AI and banking regulation by (a) providing a structured approach to quantify and optimize explainability in compliance AI systems; (b) incorporating domain-specific regulatory requirements into the mathematical framework; (c) demonstrating practical implementation through detailed constraints for each audit type; and (d) establishing a balanced approach between model performance and regulatory transparency.

The study's limitations include computational complexity in calculating Shapley values, potential oversimplification of input-output relationships through mutual information metrics, and the need for continuous recalibration as regulatory requirements evolve. Additionally, the static nature of the constraints may not fully capture the dynamic regulatory environment. Also, there is a huge computational challenge because calculating Shapley values requires examining all possible feature combinations, and the combinations grow exponentially with feature count (2^n for n features as each feature can either be included or excluded in a combination). For 50 features, that could be over a quadrillion combinations. Features such as transaction amount, frequency, location, customer risk score, merchant type, and device ID can easily add up to 50 or more. Variables like income, credit history, debt-to-income ratio, employment status, and number of credit inquiries can surpass 100. Aggregated behavioral data, such as browsing habits, transaction patterns, geolocation, and social media activity, might involve 1000+ data points. More efficient computational methods could be required, such as, sampling-based approximations of Shapley values, dimension reduction techniques, parallel computing algorithms, and feature selection to focus on most important variables (Kumar et al., 2020; Wang et al., 2024).

Future research directions should explore:

- 1) Dynamic optimization frameworks that adapt to evolving regulatory requirements
- 2) Integration of alternative explainability metrics beyond mutual information and Shapley values
- 3) Application of the model to specific use cases within each audit category
- 4) Development of more efficient computational methods for handling high-dimensional feature spaces (i.e., datasets with many variables or features)
- 5) Investigation of the model's effectiveness across different banking jurisdictions and regulatory frameworks

This research provides a foundation for developing more transparent and accountable AI systems in banking compliance. It addresses the growing need for explainable AI in regulated industries while maintaining operational efficiency.

Note: Use of Generative AI tools

During the preparation of this work, the author used Anthropic's Claude AI in order to check for spelling and grammar errors in the write-up and make corrections and to solve/check the numerical math faster. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

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